[00:00] 发言人 1: You know this analysis towards uh different assumptions they make respect to be and the third just for revision. I think you are most most familiar with small just for preparation by some of you even you use it. So intersection I made.

[00:18] 发言人 1: Give some different insights into adjustment. So now let’s first do a review. So the definition of a classification task given a feature vector remember we’re going to.

[00:40] 发言人 1: We are going to use people organizers to denote. convector. 啊, which lives in R N R N is the feature dimension。Describe describing an object typically.

[01:00] 发言人 1: Image predicts class label equal one is in class set one and always always going to use curly script letters this is curly script letters to develop a set and well why is a set of say possible classes and the definition of a. Classify learning，so given a training set，once again we use Turkish，because those are those。Sets if sign y i i from one to m m is number of training examples.

[01:49] 发言人 1: Yeah, yeah, try to say. We’re always, we’re always gonna use it. And super swift premises I to index as example reserve this. Substitute J to index the jazz feature of the ice example. Where excited is R and Y I is it increased by and goes learn class R F from R N to Y that predicts a stable overview feature vector.

[02:29] 发言人 1: And classification error rate of data set D and a function F, the classification error rate of F on D is one over M, sum from I is equal to one M. and this is the indicator function y i, this is in. 点开头方式，Y I Y E 是他的 label。Also known as ground truth label or GT label, and Fxi is a prediction of the classifier on the example, right? This is a production.

[03:22] 发言人 1: If this as event is true. 按照 indicate 方式，我给给你忘。I E V count. The number of this class say vacation examples or add by one and the sum gives you the total number of this class by examples and we divide by m gives you the rate. 可以关上第三。

[03:56] 发言人 1: The best optimal classifier uses the boolean classification function the B of X equal to R of M so R of M outputs the. In in input arguments that obtains maximal value of this function so little c from y p of y is equal to little c given x that this is posterior probability.

[04:27] 发言人 1: And is equal to arc max C to X given Y is equal to C times P Y is equal to C this is joint like food.

[04:42] 发言人 1: So don’t use simply class conditional to prediction. You have to multiply by. Class for me to give you joint joint likelihood in order to make production under the business decision。And this is equal to note nine equals.

[05:11] 发言人 1: 就是三模 log line books does it basic optimal classroom is the best classifier among all possible classifiers however it’s generally unavailable because the two class conditional distribution is unknown and it’s very difficult to estimate accurately because this is a high dimensional. Probability distribution well never gonna have sufficient data for that due to the curse of dimensionality. Right, and you

have to make approximations. And now events linear discriminants make different approximations to this as conditionals.

[06:06] 发言人 1: So last week we only deal with a single feature and how to deal with smart features. For example, this x has a dimension ninety of two and is equal to two in this case. And I use assumptions or the I use classifier approximate the basic optimal classifier.

[06:28] 发言人 1: It is approximation using a simple form of this functions. Q x given y, Q x given y. Yes. class conditional. And this formula shown that all the feature dimensions are statistically independent given the value of the class variable. conditional statistically independent so for two dimensions T X one X two given one.

[07:00] 发言人 1: is equal to show x y given one times q x two given one and generally we we may be familiar with this right p a b s two events. If they are independent, this joint probability can be factorized into the product of marginal probabilities. And in this case, we just condition those three probability on the same line, right? For that C, we condition this probability.

[07:35] 发言人 1: I C C see C A B given C is B giveC O A given times B given C that conditional and you evidence on each greater I dimension if we talk lot of bit on both. And allow us to model each dimension of the relation with a simple you are resolution here. So as you see here, if we work with a general general form, no work.

[08:11] 发言人 1: P of x one up to x ten given one，and this is called the sum iPhone one，j from one to n log p of x j given one，right? And this case we. Reduce a probability density estimation in the n dimensional space the high dimensional space to n one dimensional probability estimation problem right we just estimation one dimensional independent for.

[08:55] 发言人 1: each of ten features and the general form of the classification function. yes. Class class distribution and this class conditional due to nine best can be factored into product marginal distribution.

[09:17] 发言人 1: And we’ll consider two dimensions that introduce at the beginning of the lecture. So the x axis is pedagogy. The x axis is separate.

[09:27] 发言人 1: And we have two classes. One is with the color. The other is Virginia encoded by different colors. And we’re going to speed data set training and testing fifty percent of data for training and remaining fifty percent for testing.

[09:47] 发言人 1: I learn it we will go since the pedal length and so we are containers that we are going to using ocean distribution or so i e the gaussian. Nine base model the model each feature as a class conditional you develop ocean distribution and estimated parameters using m l e months months more that base nation. And we can visualize the two dimensional class conditionals once we have learned.

[10:21] 发言人 1: Class marginal class conditional in dimension，and we multiply together to get this。Class conditional in dimensions. And here it shows class different class conditional.

[10:36] 发言人 1: This is why it’s code one. This is why total two. Here this is the type of get other type. And as you see, for two dimensional Gaussian distribution, the shape is the level set is beliefs many large points.

[10:56] 发言人 1: On the same beliefs have the same density or really densities and inner beliefs has higher density values outer beliefs has lower density values. And you can think of as a mountain shape, alright? And the mean vector of the Gaussian distribution determines location.

[11:22] 发言人 1: Since we say two ellipses with different locations, they have different mean estimations, right? And so covariance matrix determine the. Shape of this ellipse, so they can have, can have different shapes, right? We have, have different.

[11:39] 发言人 1: Common respect estimate, however we do have a common characteristic. Between these two groups, right? So if we examine the principal axis.

[11:53] 发言人 1: Of the games it is in parallel or online with this a x access. The second principal axis of the ellipse is aligned with my axis. And this is true for all classes, right?

[12:09] 发言人 1: If we examine the principal axis of this ellipse, it’s going to be aligned with this axis. If we examine the principal axis of this ellipse, the orthogonal. Direction is going to align with the Y axis.

[12:27] 发言人 1: So why is this? Why is this true? 嗯。This because we don’t model feature correlation 为了适用，different features are conditionally independent given the class state。

[12:45] 发言人 1: Right, if they say you saw a dips something like this, it means that the first feature is positive related with the first sector. Right? So if.

[13:00] 发言人 1: We observe a larger first feature is more likely larger first feature is also large so x one predictable x two and this is not model by ibs because ibs assure statistical independence that’s why you say this. You you see the stream，本来 Will with the classic axes。And we can visualize posture.

[13:32] 发言人 1: Using the base wall and this is on training data this is on test data we delete the miss class data using the squares and here the dash black eye is a decision boundary as you see the decision boundary is very smooth right. And it’s kind of have quadratic curvature. It’s a quadratic curve.

[13:58] 发言人 1: 啊，就拿得起。Dig into the geometric interpretation of the decision boundary. Remember, the decision rule is best for this vector classification problem is our maximum from one and two, one for class one, two for class two.

[14:19] 发言人 1: One is big one equal to C and the product is equal to one to two till exchange. Given why is closing. And we take log of it, the log of product becomes the log of product becomes a sum of dogs.

[14:40] 发言人 1: And for each which I mentioned, we model it as a Gaussian distribution, right? And we can write down the Gaussian distribution. P x j given y is equal to.

[14:54] 发言人 1: C is equal to one over square root to high sigma squares，and we put。接一，把这里作为 subscript，因为 sigma is class condition and feature dependent。哎，我，I use word work here is just for rotation convenience or consistency.

[15:20] 发言人 1: You might as well delete this. 你还是说，你知道，这是 Sigma squares j comma c。It’s up to you.

[15:30] 发言人 1: You can use j commas. right. And this finish the this is function x function of.

[15:42] 发言人 1: negative x j minus mu j c squares. divide by two sigma squares j c. So this is the overall message for your ambition. If we take look on both sides. Then the right outside becomes.

[16:11] 发言人 1: 嗯，然后。One over square root two times sigma squares j d c and this no x function cancels because they are inverse of each other and this gives you negative. X J minus Mil J this words two sigma squares JC and. When you recover the second term. Here right for the first term.

[16:50] 发言人 1: We still need to do it is without to break manipulation number with here。So the log this can be written as one over two pi sigma squares j c square root can be written as to the power of one over two one half right. Is this clear and this is equal to not two pi sigma squares j c negative one over two because in the denominator we put in the numerator at another sign and we can take the x plane out of that note due to the local.

[17:42] 发言人 1: Overly，they can not sneaked over two long two 派 sigma squares JC, and that’s exactly this term，right? So why there is a left one over two because they have a score root in the denominator. That’s why. So any question this time?

[18:18] 发言人 1: Then the decision boundary consider the set of points where the log likelihood for the class one is equal to the log likelihood in class two, right? And if we move this to the left hand side. And plug into the Gaussian density function, we will give you, we will derive something like this.

[18:49] 发言人 1: And you can expanding the square terms. You can expanding the square terms here. And collecting like powers of x ones too，and collecting common terms gives you exactly a quadratic formula here。

[19:12] 发言人 1: Therefore, high quality is generally producer provide this in boundary in the two dimensional case. And this is the function of the decision operator. That’s why you see this curve. And the same is why we use this word general.

[19:37] 发言人 1: IS GENERALLY PRODUCE A RED BUSINESS ENVIRONMENT。It may it may produce minor minor is the problem right if correct terms disappears if. A one is. If a one is equal to a two for binary classification is equal to zero, and this needs to be in there. 第四项帮助。

[20:17] 发言人 1: And in LBA, you will you will see clearly when this happens. So when it arrived as the same the inner situation because the two have the same variance estimation or the generally the covariance matrix are identical so they say score that

terms cancel with each other. And have live base for boolean variable model each feature independent the absence or presence of a feature x j or input input example here the absence.

[21:00] 发言人 1: Does not mean the feature is missing. It simply simply means this feature takes the state of zero. This is equal to zero.

[21:10] 发言人 1: Note is missing. Present simply means x j is equal to one. It can take two states, either zero. And one, and for present p of x c equal to one given y equal to c, we can denote as the primary meter, this is the prime product going to estimate.

[21:32] 发言人 1: 啊，wifi jc and absence 叫 exchange，就是给 one with one minus wifi jc，the two parties should sound up one right？And my question is how are you able to. unify those two expression into one. So what’s the unify expression for p of x j is equal to y is equal to c?

[22:10] 发言人 1: This is equal to wifi j c to the power of indicator function x j is equal to one times one minus wifi j c to the power of indicator function x j is equal to zero, right? E 和 C 以及和图案。The first term survives and the second term reduces to one.

[22:40] 发言人 1: If x j is equal to zero, the first term reduces to one, the second term survives. So this is how we combine two expression one using this unified performance. You may come up with your own unified.

[23:00] 发言人 1: Equation something like this, so I can just craft it for you. Maybe indicator function one. Indicator function x j is equal to one times wifi j c plus here indication function x j is equal to zero times one minus wifi j c right once again.

[23:31] 发言人 1: And see if you want the first terms selects the second with zero because we we sounded together. And vice versa, and which which formula we want to work with? Symbolically, so the marked audio formula and additive formula.

[23:53] 发言人 1: 就是说，如果你不是 AI 模型，你还没有的。突然。And we have mle primary estimate by by j c is equal to m j c by by m c where m j c is the number of example in the class c for which j c is equal to one m c is the number of example in class c and this pictorially this is easy to understand.

[24:25] 发言人 1: So we have the total data set. And from the test set we are. Identify a subset in this subset or class label Y I is equal to C.

[24:44] 发言人 1: From this subset, we further identify a third subset, and this is where x j is equal to one. And the wifi j c is simply the number of examples in this set divided by the number of examples. in this. 所以这是第二。

[25:21] 发言人 1: And a show conditional independence class conditional distribution. fat pressing to the product margins and if you take look on both sides. The local product becomes a sum of all and for official x note for vital features present given y is equal to c and here peasant simply means.

[25:45] 发言人 1: 我这边生活里面是。X J 结构图案，哎。So smoothing some feature may not be present. 你赶紧。twenty example some pressures may not be impressing in any of twenty families that for all twenty examples you have you always gonna have exchange go to zero you don’t explain you don’t observe if the example is exited

one. And in this case, mgc is equal zero, that’s why s that’s your count, and wifi jc is equal zero accordingly.

[26:27] 发言人 1: And example in this class definitely will not come in the future, or not not come in the future means will not have exchange equal to one. And they can’t be a problem simply simply since we simply may not have seen example with a feature due to the limits that train set right if we increase expand our train set we may there there are chances that we can have examples if this put one. Therefore, setting the probability to exactly zero is kind of dangerous in theory or in any practice.

[27:06] 发言人 1: Then we have a smooth, and then we have a smooth prime alpha. I adding a virtual count in each dimension for this virtual count x j is equal to one therefore prime estimate phi phi j c is equal to m j c plus alpha because for this case dimension we add our virtual count. And we add a total of n times alpha equal to n dimensions. If r pi equal to one, this could not pass smoothing or additive smoothing. And I going to experience this effect of smoothing into this tutorial.

[27:52] 发言人 1: And you know recognizing. Smoothing the estimate helps prevents overfitting numbers. And we have a multinomial naive. What if X has a final set of categories? The XJ is in per script XJ, which has one to up to.

[28:22] 发言人 1: The category of the set instead of only two states of absence and presence variety to zero one. And here we’re going to give you a concrete example. And here’s the feature vector we are working with, right? So for the first.

[28:41] 发言人 1: Future in in we can assume it continues and we can use ocean. To model it right for x two。if is.

[28:57] 发言人 1: 功能除非看又是不 ruling. Tomorrow is and if this the surface is categoric or normal. Now, for example, you can take from A, B, C.

[29:15] 发言人 1: We can use categoric or most normal district model it. And for the fourth feature, it’s may also be a kind of work that can take on different manners. For example, it can take on one, two, three, four instead of three categories, right? That’s why we make.

[29:37] 发言人 1: the number of. categories specific to the feature dimension. We put a sub school here because different feature dimensions have different number of number of categories. So it is here.

[30:00] 发言人 1: We use a multinomial also called categorical distribution as a class conditional. So that sj is equal to x, you know, y is equal to c, is equal to y y x j c. Here we add a further subscript x here.

[30:19] 发言人 1: Because in the previous case, so we have only take zero one right? So the probability sum up one, so we may not put zero one in a subject. And this since it’s not since we have multiple choices, we must put x in the subject here as well.

[30:38] 发言人 1: And the sound. Over x is equal to one, and once again the MS estimate is MJC divide by MC. So MJC is a number example in a circle which x j equals x, right? Exactly the same.

[30:57] 发言人 1: This is D, this is subset. With when I is equal to C, this is subset. In this case, X C is equal to little X. 对，他想要拿一杯水。

[31:19] 发言人 1: All right. So now let’s move to the linear discriminant analysis. So linear discriminant already for sure is classification, a big feature that can be dated back to nineteen thirties. So it can be interpreted as a different approximation to the base optimal classifier for real valued data. Or for continuous data.

[31:48] 发言人 1: 啊，后来一比 S V 看 model，every category call and containers，and for linear is unit that is is special than for containers real wide。And so a product of independent options in my best. So LDA assumes that this class conditional follows a multi option distribution with a shared matrix.

[32:22] 发言人 1: What does that mean? So the class conditional is. Class condition, right? Mini life for each class we’re going to have a mean vector estimate.

[32:39] 发言人 1: That’s why we put a subscript C here, but for all classes we’re going to share the same priority. That’s why we don’t put a subscript C here. And this is a formula. Let’s see the formula co-variety density function for the multivariate Gaussian distribution.